Please submit the filled template in PDF (up to 2 pages) via PaperPlaza for Humanoids 2026.

Humanoid Soccer Robots (HSR)

Type and Duration

Type: Workshop
Duration: Full day

List of Organizers

Provide complete name, affiliation, e-mail. Please identify the primary contact person.
Alessandra Rossi, University of Naples Federico II and University of Hertfordshire [primary]
Vincenzo Suriani, Sapienza University of Rome [primary]
Luca Iocchi, Sapienza University of Rome

Objective and Scope

Robot soccer is a uniquely demanding benchmark for humanoid robotics: a robot must perceive and localize while moving, maintain balance under contact and uncertainty, execute agile whole-body skills, recover from falls, reason over long horizons, communicate with teammates, and make decisions in a fast, partially observable, adversarial environment. This workshop will connect the RoboCup community with the broader humanoid robotics, robot learning, control, and embodied-AI communities to examine how these capabilities can be advanced jointly rather than as isolated modules. The scope spans hardware-aware locomotion and manipulation, perception and state estimation, sim-to-real learning, foundation and vision-language-action models, multi-robot coordination, safety, benchmarking, and reproducibility. Particular attention will be given to methods that transfer beyond soccer to human-centered applications in warehouses, healthcare, inspection, entertainment, and everyday environments. The program will combine invited talks, peer-reviewed extended abstracts, short oral presentations, a panel, and an interactive roadmap session. Subject to venue and safety approval, teams will be invited to bring robots for live demonstrations or recorded demo presentations. The intended outcome is a shared set of research priorities, benchmark needs, and open challenges supporting measurable progress toward the RoboCup 2050 goal and robust general-purpose humanoids.

List of Topics

Provide a list of topics (keywords) addressed in your workshop/tutorial.
-*Autonomous humanoid soccer players*, including visual perception, localization, game understanding, and autonomous gameplay.
- *Humanoid control for robot soccer*, covering dynamic locomotion, whole-body control, physical interaction (and Safety), and robust execution under real-time constraints.
- *Foundation models for humanoid* soccer robots, including vision-language-action models, multimodal learning, large-scale robot learning, transfer learning, world models, and general-purpose embodied policies.
- *Multi-agent communication*, combining predefined communication protocols, learned communication, semantic representations, and implicit coordination.
- *Humanoid robot soccer as a benchmark for embodied intelligence*, investigating how robotic soccer competitions can support reproducible evaluation, shared datasets, standardized tasks, sim-to-real transfer, and the development of increasingly autonomous and general-purpose humanoid robots.

Expected attendance

Provide an estimate on the expected attendance to your workshop/tutorial based on the attendance of similar past events, if any.

The workshop is open to a broad audience from academia and industry researching in robotics, AI, RoboCup, humanoid robots in sports, hybrid multi-agent interaction and communication. It aims to bring together experts in robotics, artificial intelligence, machine learning, control systems, multi-agent coordination, and embodied intelligence to discuss the latest advances in autonomous humanoid systems. By fostering interdisciplinary exchange across these research areas, the workshop seeks to promote a shared understanding of how next-generation humanoid robots can leverage foundation models, robust control architectures, collaborative decision-making, and advanced communication strategies to operate effectively in dynamic, real-world environments ranging from competitive sports to broader human-centered applications. Therefore, we expect that the workshop will gain the attention of numerous attendees, with an estimated attendance of around 50 participants.

Plan to stimulate participation

We commit to promoting the visibility of the workshop through the most popular channels to reach an appropriate audience, such as mailing lists (e.g., AAAI, ACM SIGCHI, IEEE RAS), academic social networks (e.g., LinkedIn), institutional channels, and direct invitations to leading researchers in the robotics and AI.
We have also created a website (*in draft*) for sharing relevant info and updates about the workshop (e.g., program, speakers). The Workshop's organisers will also share info and updates on their own social media accounts (i.e., Instagram, LinkedIn, Facebook).
Upon acceptance, building on the strong impact and visibility of recent contributions in this area (e.g., Rossi et al., 2024), we will create a special issue in a high rank journal - such as Autonomous Robots - dedicated to the research presented at the workshop, with the goal of further disseminating and consolidating the advances discussed during the event.

Statement of inclusion, diversity, and equity

The workshop aims to foster an inclusive and diverse research community. The organizing team brings together researchers with balanced gender representation, varied scientific expertise, and a range of career stages. The organizers are committed to creating a welcoming and supportive environment that encourages participation from individuals with diverse ethnic, cultural, and professional backgrounds, as well as different levels of seniority and identities.
The workshop’s website will collect relevant material, including abstracts of the accepted contributions and info about the workshop. Our website, which will be optimised for different devices, will cater for diverse affordances in the attendees’ working environments. Keynote talks, presentations, and discussions will be shared on the website and dedicated platform (e.g., Discord), at the beginning of the conference for allowing asynchronous access and participation, and engaging with the workshop content flexibly.

Invited Speakers

Confirmed:
Alessandra Rossi, University of Naples Federico II and University of Hertfordshire.

Invited / confirmation pending:
Peter Stone, The University of Texas at Austin and Sony AI (confirmed);
Joydeep Biswas, The University of Texas at Austin (confirmed);
Sven Behnke, University of Bonn;
Changjiu Zhou, Singapore Polytechnic;

Ben Moran and Guy Lever, Google DeepMind (joint talk).

Tentative Programme

Time	Session	Lead / format
08:45-09:00	Registration and informal networking	
09:00-09:10	Welcome, objectives, and workshop challenge statement	Organizers
09:10-09:50	Keynote: Robot soccer as a grand challenge for learning, multi-agent systems, and embodied AI	**Peter Stone/Joydeep Biswas**
09:50-10:25	Contributed papers I: control, locomotion, perception, and sim-to-real	
10:25-10:50	Coffee break	
10:50-11:30		
11:30-12:25		4 lightning talks + discussion
12:25-13:40	Lunch break	
13:40-14:20	Invited talk:	Changjiu Zhou
14:20-14:55	Invited talk:	Alessandra Rossi
14:55-15:35	Joint invited talk: Learning agile robot soccer with deep reinforcement learning	Ben Moran and Guy Lever
15:35-16:15	Coffee break, posters, and robot/demo session	Authors and participating teams
16:15-16:50	Contributed papers II: foundation models, strategy, communication, and benchmarks	3 lightning talks + discussion
16:50-17:30	Panel: What must be solved next for robust humanoid soccer and general-purpose humanoids? **NVIDIA**	Industrial representative: **Yan Chang**
17:30-17:50	Interactive roadmap, takeaways, and closing	Organizers and participants

Notes & Requests

Provide a list of particular requests, i.e. poster board, extra microphones, power. Ect.